Exp-8 -Record-THRESHOLDING

Name : Mahalakshmi M

Reg No : 212224230148

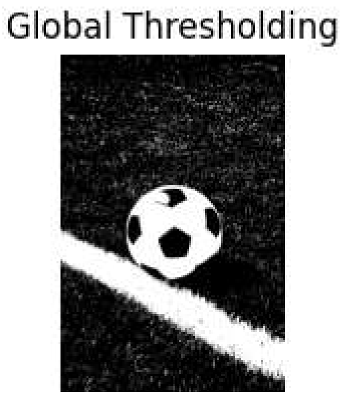
In [2]:
```
# Load the necessary packages

import cv2
import matplotlib.pyplot as plt

# Read the Image and convert to grayscale

image=cv2.imread('ball.jpg')
gray_img=cv2.cvtColor(image,cv2.COLOR_BGR2GRAY)
```

In [3]:
```
# Original image

plt.subplot(2,2,1)
plt.imshow(cv2.cvtColor(image,cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')

# Use Global thresholding to segment the image

_,global_thresholded = cv2.threshold(gray_img, 127, 255, cv2.THRESH_BINARY)

# Use Adaptive thresholding to segment the image

adaptive_thresholded = cv2.adaptiveThreshold(gray_img, 255, cv2.ADAPTIVE_THRES

# Use Otsu's method to segment the image

_,otsu_thresholded = cv2.threshold(gray_img, 0, 255, cv2.THRESH_BINARY + cv2.T
```

Original Image

In [4]:
```
# Global Thresholding
plt.subplot(2, 2, 2)
plt.imshow(global_thresholded, cmap='gray')
plt.title("Global Thresholding")
plt.axis('off')
```

Out[4]: (-0.5, 727.5, 1091.5, -0.5)



In [5]:
```
# Adaptive Thresholding
plt.subplot(2, 2, 3)
plt.imshow(adaptive_thresholded, cmap='gray')
plt.title("Adaptive Thresholding")
plt.axis('off')
```

Out[5]: (-0.5, 727.5, 1091.5, -0.5)

In [8]:

```
# Otsu's Method
plt.subplot(2, 2, 4)
plt.imshow(otsu_thresholded, cmap='gray')
plt.title("Otsu's Method")
plt.axis('off')

# Show the plot
plt.tight_layout()
plt.show()
```